Abdallah Ahmed

Alexandria, EG

+20 1050018069 • 3bdallahhahmed@gmail.com • Abdallah Ahmed

Summary

Driven Mechatronics Engineering student at the Egypt-Japan University of Science and Technology with a strong focus on integrating electronic hardware with functional robotics. I specialize in bridging the gap between core electronic component design and autonomous mechanical systems. My technical background spans end-to-end PCB design (Altium Designer), sensor integration, and mechanical prototyping. I am passionate about leveraging this multidisciplinary expertise to build reliable, high-impact engineering solutions from the ground up.

Education

Egypt-Japan University of Science and Technology — **Alexandria, EG**
Bachelor of Engineering: Mechatronics Engineering — *2024–2028*
Expected Graduation: June 2028

AlNoor International School — **Sharjah, UAE**
Secondary School Certificate (MoE Curriculum) — *2024*

Technical Experience

Egyptian Space Agency (EgSA) — **Cairo, EG**
PCB Design Trainee — *Jan 2026–Feb 2026*

- Completed an intensive 60-hour hardware design workshop, mastering the professional PCB lifecycle from core electronics theory to manufacturing workflows.
- Applied industry-standard Design Rule Checks (DRC) and prepared comprehensive documentation for fabrication and assembly.
- Gained hands-on experience in electronic component selection, hardware debugging, and design review practices.

Projects

Academic Project (— **Alexandria, EG**
Hexapod Walking Robot — *May 2026*

- Designing a biologically inspired hexapod walking robot, focusing on complex mechanical linkages and stable locomotion.
- Utilizing **SolidWorks** for 3D modeling and structural design, preparing optimized parts for rapid prototyping via 3D printing.

EgSA Workshop Project — **Cairo, EG**
End-to-End PCB Design: Arduino Uno R3 & 5V Power Supply — *Jan 2026–Feb 2026*

- Engineered production-ready PCBs for an Arduino microcontroller and a 5V power supply from scratch using **Altium Designer**.
- Developed custom component libraries, including schematic symbols and 3D footprints, to support complex board layouts and advanced routing.
- Managed the complete project lifecycle, moving successfully from initial schematic capture to a finalized production-ready layout.

SoiLAB Eco-Warrior Innovation Camp — **Sharjah, UAE**
Autonomous Air Quality Monitoring Robot (Segway) — *Aug 2023–Sep 2023*

- Developed a functional Segway robotics platform integrated with environmental sensors to detect and analyze pollution levels in real-time.
- Engineered a control system to regulate air quality data collection, demonstrating the fusion of mobile robotics and environmental technology.
- Utilized advanced fabrication facilities for hardware chassis prototyping and systems integration.
- **Sustainability Focus:** Applied Mechatronics principles to address urban air quality.

Key Skills

Engineering: Robotics, Sensor Integration, Prototyping, Control Systems, Schematic Capture

Tools: Altium Designer, Full PCB Design Lifecycle, Custom Component Libraries, End-to-End Solutions

Soft Skills: Analytical Thinking, Systems Thinking, Effective Communication, Project Management

Languages: Arabic (Native), English (Advanced Proficiency), Japanese (Beginner)

Certifications

2026: Comprehensive PCB Design Workshop - Egyptian Space Agency